

Heart Disease Prediction Using Neural Networks and TensorFlow

A Deep Learning Approach for Early Cardiovascular Disease Detection

Abstract

Heart disease is one of the leading causes of death worldwide. Early detection and diagnosis can significantly improve patient outcomes and reduce mortality rates. This project presents a Heart Disease Prediction System built using an Artificial Neural Network (ANN) implemented with TensorFlow. The model analyzes a range of patient health attributes and predicts whether a patient is likely to have heart disease. The dataset was preprocessed, scaled, and split into training and testing sets before being used to train the network. The final model — a four-layer architecture with a Dropout regularization layer — achieved a test accuracy of **95.12%**, with precision, recall, and F1-score all at approximately **0.95** across both classes, confirming strong and balanced predictive performance.

Keywords: Neural Networks, TensorFlow, Machine Learning, Heart Disease Prediction, Classification, Artificial Intelligence, Dropout Regularization

1. Introduction

Heart disease remains one of the most serious health concerns worldwide. According to medical research, millions of people suffer from cardiovascular diseases every year. Early diagnosis can help healthcare professionals provide timely treatment and improve survival rates.

Machine Learning and Artificial Intelligence have become powerful tools in healthcare applications. Neural Networks, inspired by the structure of the human brain, can learn complex patterns from medical data and assist in disease prediction.

The objective of this project is to develop a neural network-based model capable of predicting heart disease using patient health information. TensorFlow is used as the primary deep learning framework for model development, with a Dropout layer incorporated to reduce overfitting and improve generalization on unseen patient data.

2. Problem Statement

The goal of this project is to build a classification model that can determine whether a patient has heart disease based on various medical attributes such as age, blood pressure, cholesterol levels, and other health indicators.

Objectives

- Analyze the Heart Disease dataset.

- Preprocess and prepare the data for training.
- Develop a Neural Network using TensorFlow with Dropout regularization.
- Train and evaluate the model.
- Predict heart disease for new patient records.

3. Dataset Description

The Heart Disease Dataset contains medical information collected from patients.

Features Included

Feature	Description
Age	Age of the patient
Sex	Gender of the patient
Chest Pain Type	Type of chest pain experienced
Resting Blood Pressure	Blood pressure while resting
Cholesterol	Serum cholesterol level
Fasting Blood Sugar	Blood sugar level
Resting ECG	Electrocardiographic results
Maximum Heart Rate	Maximum heart rate achieved
Exercise Induced Angina	Exercise-related chest pain
ST Depression	Depression induced by exercise
Slope	Slope of peak exercise ST segment
Number of Vessels	Number of major vessels colored by fluoroscopy
Thalassemia	Blood disorder information
Target	Presence or absence of heart disease

Target Variable

- **0** = No Heart Disease
- **1** = Heart Disease Present

4. Tools and Technologies

The following technologies were used during project development:

- Python

- TensorFlow
- Keras
- Pandas
- NumPy
- Scikit-Learn
- Matplotlib
- Jupyter Notebook

5. Methodology

The project follows a standard Machine Learning workflow.

Step 1: Data Collection

The Heart Disease dataset is loaded using Pandas.

Step 2: Data Exploration

The dataset is examined to understand:

- Number of records
- Feature types
- Missing values
- Statistical information

Step 3: Data Preprocessing

Data preprocessing includes:

- Handling missing values (if any)
- Separating features and target variable
- Scaling features using StandardScaler

Feature scaling ensures that all features contribute equally during training.

Step 4: Train-Test Split

The dataset is divided into:

- Training Data: 80%
- Testing Data: 20% (205 samples used for final evaluation)

This allows the model to learn from one portion of the data and be evaluated on unseen data.

6. Neural Network Architecture

The final Artificial Neural Network consists of an input layer, two hidden dense layers separated by a Dropout regularization layer, and a sigmoid output layer:

Layer	Units / Rate	Activation
Hidden Layer 1 (Dense)	64 Neurons	ReLU
Dropout Layer	Rate = 0.1	—
Hidden Layer 2 (Dense)	32 Neurons	ReLU
Output Layer (Dense)	1 Neuron	Sigmoid

The architecture can be represented as:

```
Input Layer → Dense(64, ReLU) → Dropout(0.1) → Dense(32, ReLU) → Dense(1, Sigmoid)
```

The Dropout layer (rate = 0.1) randomly deactivates 10% of neuron connections during training, which helps prevent the network from overfitting to the training data and improves its ability to generalize to new, unseen patients.

7. Model Training

The model is compiled using:

- **Optimizer:** Adam Optimizer
- **Loss Function:** Binary Crossentropy
- **Evaluation Metric:** Accuracy

Training Parameters:

- Epochs: 50
- Batch Size: 16
- Validation Split: 20%

The training process allows the model to learn patterns from the medical data and adjust its weights through backpropagation.

8. Results and Evaluation

The model is evaluated using the testing dataset (205 samples).

Overall Test Accuracy

The final trained model achieved a test accuracy of **95.12%** (0.9512194991111755).

Classification Report

Class	Precision	Recall	F1-Score	Support
0 (No Disease)	0.93	0.97	0.95	102
1 (Disease Present)	0.97	0.93	0.95	103
Accuracy			0.95	205
Macro Avg	0.95	0.95	0.95	205
Weighted Avg	0.95	0.95	0.95	205

Confusion Matrix

Of the 205 test samples, the model correctly classified 99 of 102 negative cases and 96 of 103 positive cases, with only 3 false positives and 7 false negatives.

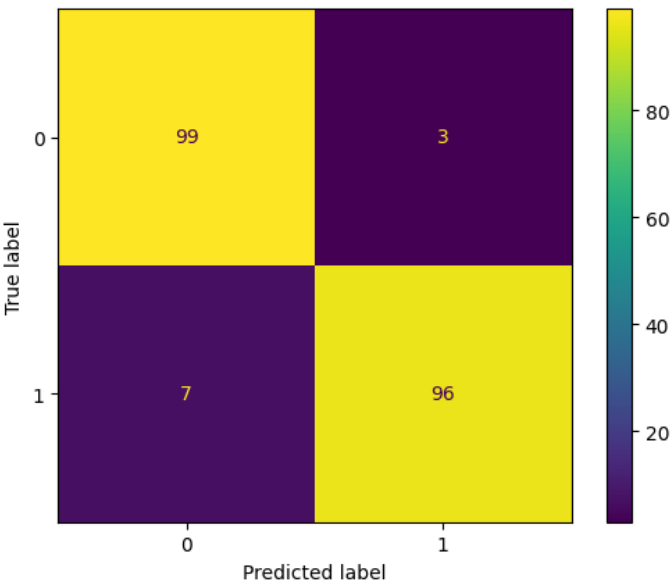


Figure 1: Confusion matrix of the model on the test set.

Training vs. Validation Accuracy

The training and validation accuracy curves over 50 epochs show steady improvement, with training accuracy approaching 0.98–0.99 and validation accuracy stabilizing around 0.93–0.95, indicating good convergence with limited overfitting due to the Dropout layer.

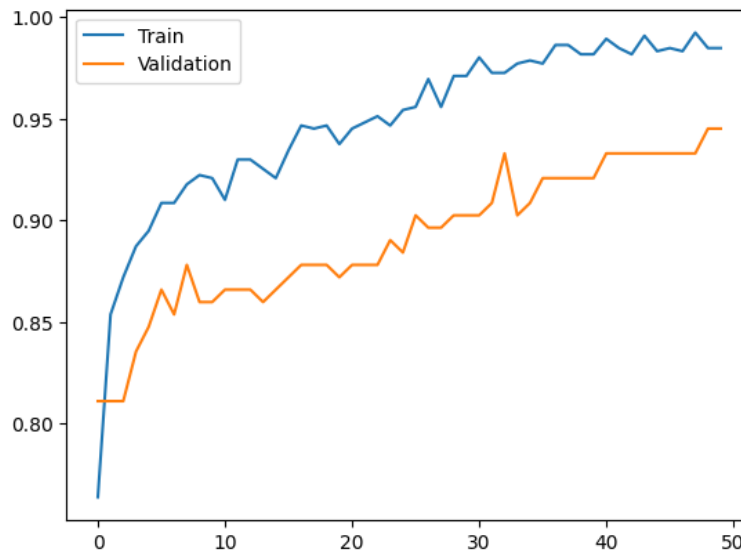


Figure 2: Training and validation accuracy over 50 epochs.

9. Discussion

The Neural Network successfully learns patterns from patient health records and predicts the likelihood of heart disease with high reliability. The use of feature scaling improves model convergence, while the Dropout layer (rate = 0.1) reduces overfitting and contributes to the model's strong generalization performance.

With a test accuracy of 95.12% and balanced precision, recall, and F1-scores (~0.95) across both classes, the results demonstrate that Artificial Neural Networks can be effectively applied to healthcare prediction problems. The confusion matrix confirms that misclassifications are low and roughly balanced between false positives and false negatives, which is important in a medical screening context where both error types carry clinical consequences.

However, model performance still depends heavily on data quality, dataset size, and parameter tuning. Potential improvements include:

- Increasing dataset size
- Further hyperparameter tuning
- Experimenting with additional Dropout layers or rates
- Using deeper architectures
- Applying Cross Validation

10. Conclusion

This project demonstrates the implementation of a Heart Disease Prediction System using TensorFlow and an Artificial Neural Network with Dropout regularization. The model processes patient health information and predicts the presence of heart disease with a strong test accuracy of 95.12% and balanced precision/recall across both classes.

The project highlights the potential of Artificial Intelligence in assisting healthcare professionals with early disease detection and decision-making. Future work can focus on further improving model performance and deploying the system as a web-based healthcare application.

References

- [TensorFlow Documentation](#)
- [Scikit-Learn Documentation](#)
- [Keras Documentation](#)
- [UCI Machine Learning Repository](#)
- [Heart Disease Dataset Documentation](#)